

quency. Known as noise jamming, this prevents the enemy radar from effectively executing its job.

There are many types of jamming that are carried out. Jamming techniques fall outside the purview of this article. Radio communications can also be effectively jammed apart from radar jamming.

**Spoofing.** This can be done electronically and also by other means. That is, to make the enemy radar see a target where actually there is none and also to make the radar not see the target where it actually is. Electronic spoofing methodologies fall out of the purview of this article.

Another effective means of spoofing is through chaff. Tiny, weightless metal foils like the husk of food grains are called chaff. Chaff is dumped in thousands in a shell and fired through mini cannons in the path of the radar of, say, an incoming missile. The shell explodes high up in the air and lets the chaff out. Length of the chaff is half the wavelength of the radar for better signal reflection. Being very thin and light, these foils stay in the air like a cloud and take time to settle down. So in this intervening time, these effectively reflect radar energy.

Simultaneously, the ship rapidly changes its course. The distance and direction of the chaff shell firing is controlled by EW systems. By skillfully firing chaff in the path of the cruise missile, a carrier has a fair chance to mislead the cruise missile to fly away from its course.

**Airborne EW systems.** Apart from the built-in ESM and ECM systems, the CSG can employ EW aircraft called EA-18 Growlers. These aircraft have a variety of jamming systems and can fly to an area according to the requirement and jam the enemy radar and communication systems.

Typically, when combat aircraft are sent to strike ground targets deep inside enemy territory, the targets are heavily defended by surface-to-air-missile (SAM) systems. It is these EW aircraft that first enter the area

and perform jamming to protect the strike aircraft from SAM systems. These aircraft are also capable of gathering signal intelligence by gathering enemy signal samples. Airborne EW systems also work in conjunction with ship-borne EW modules.

## Under-sea warfare

A threat for CSG is the presence of an enemy submarine. For a submarine, the prize target is the carrier of the CSG. Submarines use anti-ship cruise missiles for long-range engagement and torpedoes for close-range engagements. Modern-day nuclear submarines can submerge at their will for months together.

For a CSG, the first line of defence against submarines is formed by associated nuclear submarines. These submarines sail far ahead of the CSG and look for enemy submarines. Ocean surfaces are not plain; there are gorges and canyons under the sea. Such oceanographic features are exploited by enemy submarines. Submarines of the CSG look for the enemy submarines at these points. If they find one, then and there it is engaged.

The second line of defence is formed by associated Aegis destroyers. These have both active and passive sonars mounted on their respective hulls. These sonars are called hull-mounted sonars.

An active sonar

emits Ping sounds and looks for echoes. Active sonars are complemented by passive sonars. Passive sonars only listen to underwater sound. Unbelievably, sound waves travel enormous distances due to convergence zones in oceans. These zones occur as concentric circles for every 61km from the source of sound. Experienced passive sonar operators can detect even the sound resulting due to a sailor accidentally dropping a tool in the submarine.

When an enemy submarine is detected, the destroyer may engage it with torpedoes for close-range engagement and a missile-torpedo combo for long-range engagement. The latter is a missile with a torpedo as a warhead. If the submarine's presence is detected at a fairly longer range, this missile takes the torpedo near the target area where the torpedo jettisons itself and dives into the sea. From there the torpedo continues the attack.

Further, to increase the range of submarine detection, onboard under-sea warfare helicopters called Sea



During the Falklands war between the UK and Argentina, British nuclear submarine HMS Conqueror launched three torpedoes and sank Argentine cruiser ARA General Belgrano. As an aftermath, Argentine carrier ARA Veinticinco de Mayo was sailed back to the harbour to protect it from the British submarine. In the 1971 India-Pakistan War, the lone Indian carrier INS Vikrant had to be anchored near Andaman Islands to protect it from Pakistani submarine PNS Ghazi. INS Vikrant came to action only after PNS Ghazi was lured and destroyed outside Vizag harbour